Revision Exercise
Review Questions

Total points: 100

The **Fruit Sales Management Application** is designed to help a fruit shop owner manage fruit products and their expiry dates. Users can view the list of all fruits, add a new fruit, update the expiry date of a fruit, delete a fruit, and view fruits that are about to expire. All fruits are persisted using a SQLite database.

Data Structure

Each fruit is stored in a SQLite table named `fruits` with the following fields:

- `id INTEGER PRIMARY KEY AUTOINCREMENT`: the identifier of the fruit (e.g., 1, 2, 3).
- `name TEXT NOT NULL`: the name of the fruit (e.g., "Xoai Cat Chu").
- `origin TEXT`: the origin of the fruit (e.g., "Dong Thap").
- `price REAL NOT NULL`: the selling price per kg in VND (e.g., 25000).
- `expiredDate TEXT NOT NULL`: the expiry date in the format `YYYY-MM-DD` (e.g., "2026-11-15").

General Requirements

- **Project folder**: must be named `fa26_prm393_StudentCode` (e.g., `fa26_prm393_ce123456`).
- **Development tool**: use Android Studio as your development IDE.
- **Technology**: use Flutter for the mobile app and SQLite for database management.
- **Data**: data must be stored and managed using SQLite.
- **State management**: the global fruit-list state must be managed using **Provider**.

Tasks

1. **List of fruits (List Screen)** Create the List Screen to display the list of fruits using a `ListView`, including: Name, Origin, Price, and Expiry date. [25 points]
   - The list must be fetched from the SQLite database. If the fruits list is empty, display a simple text instead. When the user taps the "+" `FloatingActionButton`, navigate to the Add fruit screen. (5)
   - Otherwise, display the list using a `ListView` and each fruit in a `ListTile`. The `ListTile` has `Icons.eco` as a leading icon, the fruit name as the title, and "Origin – Price – Expiry date" as the subtitle. Provide `Icons.edit` and `Icons.delete` as the trailing actions (Edit follows Task 3; Delete follows Task 4). (20)
2. **Add New Fruit (Add Screen)** Create the Add Screen to display a Form to add a new fruit, including: Name, Origin, Price, Expiry date. [20 points]
   - The FormField validates that the input data is required before submitting the form. (5)
   - The price FormField uses the number keyboard type. (5)
   - The expiry date is chosen with `showDatePicker` and stored as `YYYY-MM-DD`. A status badge (Fresh / Expiring soon / Expired) updates dynamically as the date changes. (5)
   - After the user taps the "Save" button, the new fruit must be saved in the SQLite database and the application routes the user back to the List Screen. The List Screen must immediately display the updated data without restarting the application. (5)

3. **Update expiry date (Edit)** When the user taps the `Icons.edit` icon of a fruit, open a form to update the **expiry date** of that fruit. [15 points]
   - The form is pre-filled with the selected fruit's current expiry date. (5)
   - The user picks a new expiry date with `showDatePicker`. The status badge updates dynamically as the new date changes. (5)
   - After the user taps "Update", the new expiry date is saved to the SQLite database and the List Screen is refreshed. Other fields are not edited. (5)
4. **Delete a fruit** Create the Delete confirmation dialogue. [15 points]
   - After the user taps the "Delete" icon, open a dialogue to confirm the action, displaying the chosen fruit's name. (5)
   - After the user taps "Delete", the chosen fruit is removed from the SQLite database and the List Screen is refreshed. (10)
5. **Expiry Warning Screen** Add an AppBar action (`Icons.event_busy`) that opens a dedicated screen listing fruits that are about to expire or have expired. [15 points]
   - Compute `daysLeft = expiredDate.difference(today).inDays` using the device's local date. List fruits with `0 <= daysLeft <= 7` under an "Expiring soon" section. (5)
   - List fruits with `daysLeft < 0` under an "Expired" section. Show the number of fruits in each section. (5)
   - Recalculate and refresh the warning list whenever a fruit's expiry date is added or updated. (5)
6. **Correct practices with Dart/Flutter development and UI respect** [10 points]
   - Respect UI suggestions: AppBar titles, color is orange, components displayed without overflow. (5)
   - Dispose the owned controllers after use, use correct OOP, and apply consistent theming. (5)

Sample Data (Manual Testing)

Enter each record through the Add fruit screen. ID is auto-generated. Use the reference local date **2026-11-08** to verify the Expiry Warning Screen.

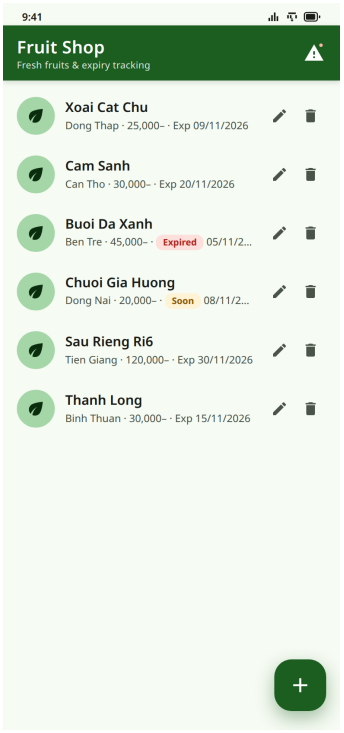
Id	Name	Origin	Price	Expiry	Expected group
1	Xoai Cat Chu	Dong Thap	25,000	2026-11-09	Expiring (1 day)
2	Cam Sanh	Can Tho	30,000	2026-11-20	Fresh
3	Buoi Da Xanh	Ben Tre	45,000	2026-11-05	Expired
4	Chuoi Gia Huong	Dong Nai	20,000	2026-11-08	Expiring (0 day)
5	Dau Tay Da Lat	Lam Dong	80,000	2026-11-14	Expiring (6 days)
6	Sau Rieng Ri6	Tien Giang	120,000	2026-11-30	Fresh
7	Nho Ninh Thuan	Ninh Thuan	65,000	2026-11-02	Expired
8	Thanh Long	Binh Thuan	30,000	2026-11-15	Expiring (7 days)
9	Mit Thai	Long An	35,000	2026-12-01	Fresh
10	Bo Sap	Dak Lak	40,000	2026-11-16	Fresh

Expected results at reference date 2026-11-08:

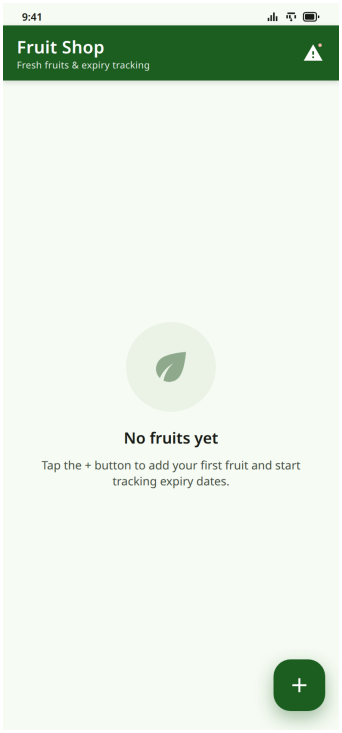
- Expired: **2** | Expiring soon ($0 \leq$ daysLeft ≤ 7): **4** | Fresh (> 7 days): **4**

Illustrations

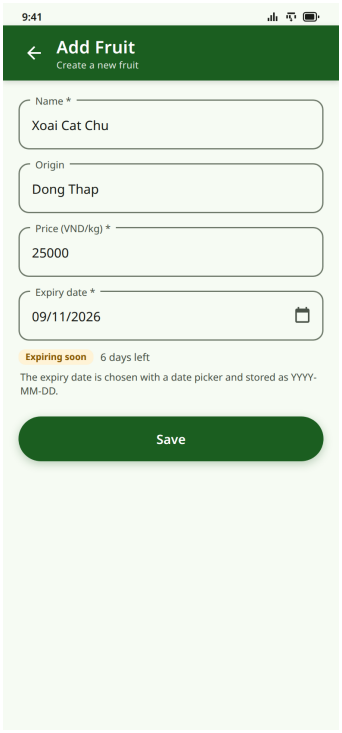
The following screens illustrate the expected Material 3 user interface for each task. Colors are dark green based on the fruit theme.



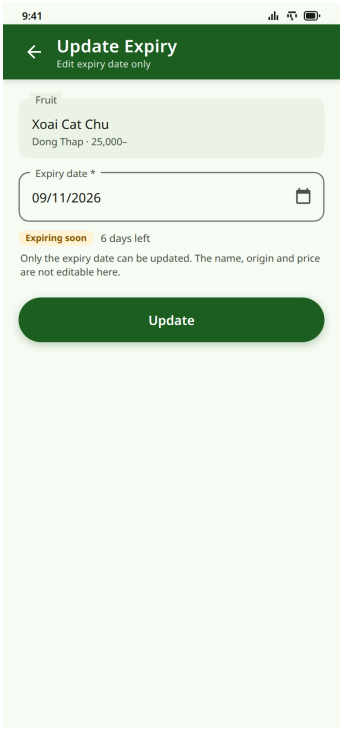
(a) Fruit List



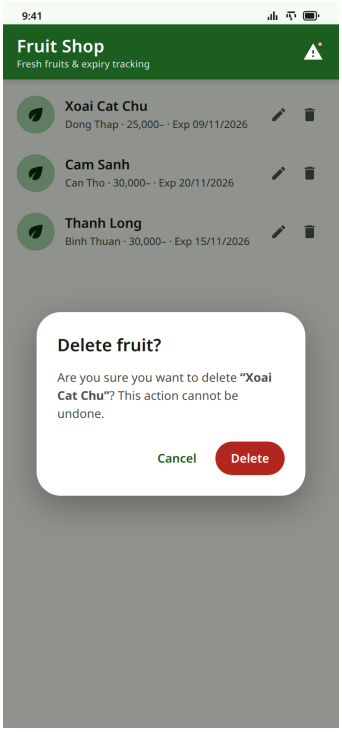
(b) Empty state



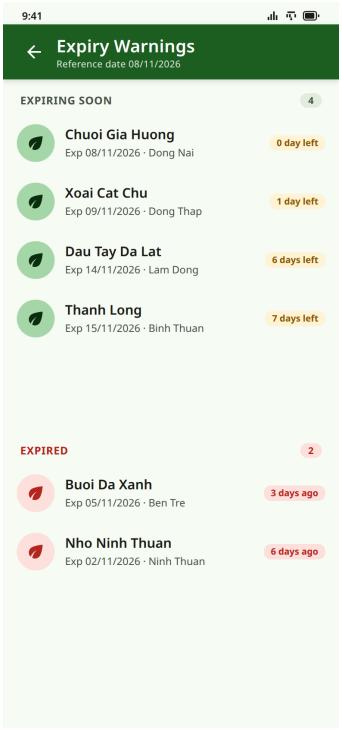
(c) Add Fruit



(d) Update Expiry



(e) Delete confirmation



(f) Expiry Warnings

- Students may use compatible Flutter/Dart versions and offline resources available on their computers.
- Run `flutter clean` before submitting the project folder. No compression is required.
- Students are **NOT ALLOWED** to use any Internet resources such as chat programs, email, remote desktop, and especially AI tools. These actions are considered cheating and will receive zero points.
- **Rules**: the project will receive zero points if it fails to run, or if it contains data structures, variables, images, screens, or UI elements unrelated to the examination topic.

— THE END —